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Bleeding rates, healthcare utilization, and costs among patients with hemophilia a without inhibitors treated with concomitant octocog alfa or extended half-life factor VIII while on emicizumab prophylaxis

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ABSTRACT

Aim: Emicizumab has changed prophylactic treatment for HA, but the need for FVIII replacement still exists for the treatment of breakthrough bleeds, perioperative management, and participation in high-risk activities/sports. The primary objectives of this study were to describe the characteristics of individuals with congenital hemophilia A (HA) without inhibitors receiving emicizumab prophylaxis and to compare clinical outcomes, healthcare resource utilization (HCRU), and costs among those prescribed octocog alfa [Advate, standard half-life factor VIII (FVIII)] *versus* extended half-life (EHL) FVIII agents.

Methods: Patients with congenital HA without inhibitors and emicizumab use were identified retrospectively from Inovalon claims data and stratified by concomitant octocog alfa *vs.* EHL FVIII use. Inverse probability treatment weighting (IPTW) and multivariable regression were used to balance baseline covariates and compare billable annualized rate of bleeding (ABR_b), healthcare utilization, and costs between subgroups.

Results: Among the 1,282 patients receiving emicizumab prophylaxis, 501 received concomitant FVIII prescription for octocog alfa ($n = 274$) or EHL ($n = 227$) during follow-up. Following IPTW adjustment, total and medical costs were comparable between groups; however, pharmacy costs associated with concomitant octocog alfa therapy were significantly lower compared with EHL products ($35,381 *vs.* $61,739; $p < 0.001$). Importantly, ABR_b (0.35 *vs.* 0.28; $p = 0.34$) and bleed-related healthcare utilization remained similar across cohorts.

Limitations: Findings are limited by the retrospective design and reliance on claims data, which may omit home-treated bleeds, and misrepresent FVIII utilization due to billing complexities and assumptions about dispensed product use.

Conclusions: While the choice of concomitant FVIII does not substantially influence ABR_b or bleed-related HCRU, significantly lower pharmacy costs with octocog alfa compared with EHL agents highlight its potential cost saving and support its consideration as a preferred on-demand treatment option in patients with HA without inhibitors on emicizumab prophylaxis.

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Introduction

Hemophilia A (HA) is an X-linked congenital bleeding disorder caused by deficiency of factor VIII (FVIII). It affects ~12 per 100,000 males in the United States and is associated with a substantial disease burden. The severity of HA is strongly correlated with FVIII activity levels, ranging from mild to severe, with the primary goal of treatment being the prevention of both acute and long-term bleeding

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complications[1–4]. HA severity is classified based on FVIII activity levels, expressed as a percentage of normal activity: mild (>5% up to 40%), moderate (1–5%), and severe (<1%)[5].

Prophylactic treatment with FVIII concentrates or non-factor therapies is the standard approach for managing patients with severe HA to prevent bleeding episodes[1,4,6]. FVIII concentrates can be classified as either plasma-derived or recombinant[7]. Recombinant FVIII (rFVIII) products are categorized according to their purity levels as first, second, and third-generation rFVIII concentrates, with each generation representing an increase in purity. rFVIII products are produced using recombinant DNA (rDNA) technology, which involves the genetic modification of host cells[8–10]. The rFVIII products include products with a standard half-life (SHL), which typically averages around 12 h, and products with an extended half-life (EHL), which ranges from 1.4 to 1.6 times longer than SHL FVIII products, or more recently, 4 times longer with efanesoctocog alfa[11]. Technologies such as Fc fusion, albumin fusion, protein modification, and PEGylation have been used to achieve the prolonged half-life in EHL products, allowing for less frequent dosing or maintaining a higher FVIII activity trough level (PROPEL)[6,12]. SHL FVIII, specifically octocog alfa (Advate), is a widely used factor replacement therapy for HA prophylaxis and in the U.S. and typically requires intravenous infusions three to four times weekly[13].

Emicizumab, the first approved non-factor therapy for HA, expanded the treatment landscape and offers an alternative to FVIII concentrates for prophylactic treatment in patients with severe HA, both with and without inhibitors[1,14,15]. This humanized, bispecific FVIII-mimetic was approved in the United States (US) for HA without inhibitors in October 2018[16,17]. Emicizumab has demonstrated a favorable safety profile and is administered subcutaneously at an interval of weekly, biweekly, or every 4 weeks[4]. It has been shown to effectively reduce bleeding episodes, preserve joint function, and lessen treatment burden.

While emicizumab has changed prophylactic management in patients with HA, studies conducted in real-world setting indicate that breakthrough bleeding still occurs, with an annualized bleeding rate of 1.4 in children and 1.6 in adults receiving emicizumab prophylaxis[18,19]. Additionally, planned or unplanned concomitant FVIII was required in 60–80% of invasive procedures among patients with HA without inhibitors receiving emicizumab prophylaxis[20,21]. In the major surgery subgroup, FVIII was administered in all procedures in this population[21]. These findings underscore the ongoing need for concomitant FVIII replacement in patients with HA without inhibitors on emicizumab prophylaxis, specifically for the treatment of breakthrough bleeds, perioperative management, and for supplemental activity-related prophylaxis[22].

Despite the current availability of 17 FVIII products on the market, including five plasma-derived SHLs, seven recombinant SHLs, and five recombinant EHLs, there is a dearth of head-to-head comparative data to determine the optimal FVIII product for on-demand treatment of breakthrough bleeds in patients receiving emicizumab prophylaxis, with most available data focusing on the safety and efficacy of emicizumab alone. This presents a gap in understanding potential differences in clinical outcomes, healthcare resource utilization (HCRU), and associated costs of concomitant use of FVIII products during emicizumab prophylaxis. Addressing this gap is essential to inform evidence-based treatment decisions, optimize patient care, and guide formulary and policy choices in the evolving landscape of HA management.

The primary objectives of this study were to describe the demographic and clinical characteristics of individuals with congenital HA without inhibitors receiving emicizumab prophylaxis and to compare billable annualized rate of bleeding (ABR_b), HCRU, and costs among those prescribed octocog alfa (Advate, SHL FVIII)[2] *versus* EHL FVIII agents. Additionally, the study seeks to explore the real-world frequency and context in which concomitant FVIII products are prescribed alongside emicizumab. To our knowledge, this is the first reported analysis comparing the aforementioned criteria across FVIII products.

Methods

Study population

This retrospective, cohort study utilized secondary closed claims data to evaluate concomitant prescription of octocog alfa *versus* EHL FVIII products in patients receiving prophylactic treatment with emicizumab. The analysis focused on clinical outcomes, HCRU, and costs.

The analysis used deidentified patient-level data from Inovalon, provided by HealthVerity, which maintains one of the largest US closed claims datasets. This dataset included longitudinal healthcare data from over 160 payer plans in the US across Medicare Fee-for-Service (FFS), Medicare Advantage, Medicaid, and Commercial insurance programs, enabling comprehensive capture of healthcare encounters.

Patients were eligible for inclusion if they had a diagnosis of congenital HA (ICD-10-CM: D66) from 1 February 2023 to 29 October 2024 (study period), at least one emicizumab claim after the study start date, and a minimum of 12 months (baseline period) of continuous enrollment before the index date. Index date was defined as the earliest emicizumab claim during the study period. Patient follow-up began on the index date and continued until the earliest of disenrollment, death, a switch in concomitant FVIII treatment, emicizumab discontinuation, or the end of the study period. Emicizumab discontinuation was defined as the last observed claim for emicizumab, or the second-to-last claim if followed by a gap of 90 days or more before the next expected dose.

Patients were excluded if there was evidence of FVIII inhibitors during the baseline period, defined by the presence of at least one claim for bypassing agents [(anti-inhibitor coagulant complex (FEIBA), coagulation factor VIIa (NovoSeven), or eptacog beta (Sevenfact)], at least one claim for rituximab, or signs of immune tolerance induction therapy, operationalized as >3 times the median IU dispensed for patients in the same age group across five consecutive 28-day intervals. Additional exclusions included patients with two or more diagnoses (at least 30 days apart) for von Willebrand disease, hemophilia B (HB), acquired HA, or acquired coagulation factor deficiency during the 12-month baseline period[23].

During the study period, patients fulfilling the inclusion criteria were classified into two concomitant FVIII exposure cohorts: the octocog alfa cohort (at least one pharmacy claim for octocog alfa (Advate) and the EHL cohort (at least one pharmacy claim for efmoroctocog alfa (Eloctate), rurioctocog alfa pegol (Adynovate), turoctocog alfa pegol (Esperoct), damoctocog alfa pegol (Jivi), or efanesoctocog alfa (Altuviiio). This patient classification into concomitant FVIII exposure groups was based solely on National Drug Codes (NDC), as Healthcare Common Coding System's J codes are not specific to octocog alfa.

A distinct efanesoctocog alfa sub-cohort was identified from within the EHL group, defined by at least one pharmacy claim for efanesoctocog alfa, to be used for descriptive purposes only. This sub-cohort was of particular interest due to the recent approval of efanesoctocog alfa in 2023 and the minimal amount of post-approval RWE or observational research compared to other products. Combined pharmacy costs, defined as expenditures related to FVIII and emicizumab prescriptions, were identified using the relevant NDC and J codes. Diagnostic costs were evaluated using medical claims data.

Study variables

Demographic and clinical characteristics

The analysis considered a comprehensive set of demographic and clinical characteristics, including age (as both a continuous variable and categorized as <6, 6–12, 12–18, and 18+ years), gender, payer type (commercial, Medicaid, Medicare Advantage), region (Northeast, Midwest, South, West), and index year (2023 or 2024). Clinical factors assessed during the baseline period included history of infections (e.g. hepatitis B/C, HIV/AIDS, sepsis, central line infections, pneumonia), metabolic conditions (diabetes, hyperlipidemia, hypothyroidism), cardiovascular conditions (e.g. hypertension, various thrombotic events, heart failure, CAD, stroke, myocardial infarction), bleeding history (major/minor), psychiatric conditions (depression, anxiety, bipolar disorder), and other comorbidities such as liver disease, osteoporosis, arthritis, arthropathy, hematuria, joint replacements, GERD, acute kidney failure, post-hemorrhagic anemia, hemarthrosis, asthma, COPD, respiratory insufficiency, and epilepsy. Mortality risk of patients was also described using the Charlson Comorbidity Index (CCI). Bleeding history was defined as at least one medical claim for a relevant bleeding event during the baseline period. Bleeding-related medical claims were classified as major or minor by a clinician (Table S5).

Annualized rate of bleeding

ABR_b was defined as the annualized rate of ICD 10 codes for bleeding events captured during the follow-up period. All bleeding codes (both inpatient and outpatient) were grouped based on the body part where the bleeding event occurred and were informed by prior studies[19,23–25]. Multiple bleeding claims for the same body part around a 7-day window were grouped together as one bleeding event[23,24]. If multiple claims for bleeds occurred in different body parts, with or without the 7-day window, they were treated as separate claims[23].

Healthcare resource utilization and cost

All-cause and bleed-related HCRU outcomes were assessed and reported as annualized means. All HCRU outcomes were categorized as inpatient visits, outpatient visits, and emergency room (ER) visits. Cost outcomes were also calculated using linked NDC and J codes and reported as annualized means to account for varying follow-up durations across patients, providing a standardized measure of treatment-related healthcare costs within the cohorts of interest. Total cost of care was classified as pharmacy costs and all-cause medical costs. Pharmacy costs represented the total of all drug and procedure related costs associated with emicizumab or FVIII products and were calculated based on the average sales price (ASP) per international unit, calculated *via* NDC and J codes, respectively. Costs related to the management of comorbidities were not included. Medical cost consisted of diagnostic cost associated with medical claim and is captured at a daily level, which may or may not have been directly related to HA. FVIII utilization was defined as the combined number of FVIII-related pharmacy and procedure claims. Of note, the HCPCS non-specific code J7192 was included in utilization calculation for patients within the sub-cohorts of octocog alfa, damoctocog alfa pegol, and turoctocog alfa pegol.

Statistical analysis

Data transformation

Outcomes in the study were annualized to account for variable follow-up durations across patients. Annualized pharmacy costs were calculated using treatment dispensation data in conjunction with ASP per unit values obtained from the Centers for Medicare & Medicaid Services (CMS)[26]. Dispensed units were extracted from source data, with outliers—defined as values outside the range of (Median ± 1.5*IQR)—standardized to the respective limits. To minimize the influence of extreme values, the top 1% of annualized dispensed units within each cohort were capped at the 99th percentile. The total dispensed units per treatment per patient were multiplied by the corresponding ASP to derive pharmacy costs, which were then annualized based on the patient-specific follow-up period. ABR, HCRU, and medical costs were similarly annualized by dividing individual totals by the follow-up period in years. Statistical analyses were conducted using these annualized values at the patient level.

Statistical methods

Descriptive statistics comprising means, *SD*s, medians, and IQRs for continuous variables, and frequencies and proportions for categorical variables, were reported for baseline demographic and clinical characteristics of study participants and for all outcomes.

Multivariable analysis was conducted using IPTW to adjust for baseline differences across concomitant octocog alfa (Advate) and EHL patient groups and create balanced pseudo-cohorts for comparison[27]. Propensity scores were estimated *via* multivariable logistic regression using covariates selected based on baseline imbalances. Other criteria for covariate selection included clinical relevance and certain association with exposure based on literature support, and statistical criteria—specifically, variables with a minimum frequency of 15 across treatment groups and low multicollinearity (Pearson correlation $\leq$ 0.8). A threshold of 15 patients was chosen based on a general rule of thumb that allows roughly 6–10 patients per covariate included[28]. Furthermore, too many covariates may compromise the accuracy and precision of propensity scores[29]. Normalized IPTW weights were calculated using propensity scores. Covariate balance between cohorts before and after weighting was assessed using standardized mean differences and distributional comparisons (Figure 1). All IPTW-based analyses were conducted using the R package

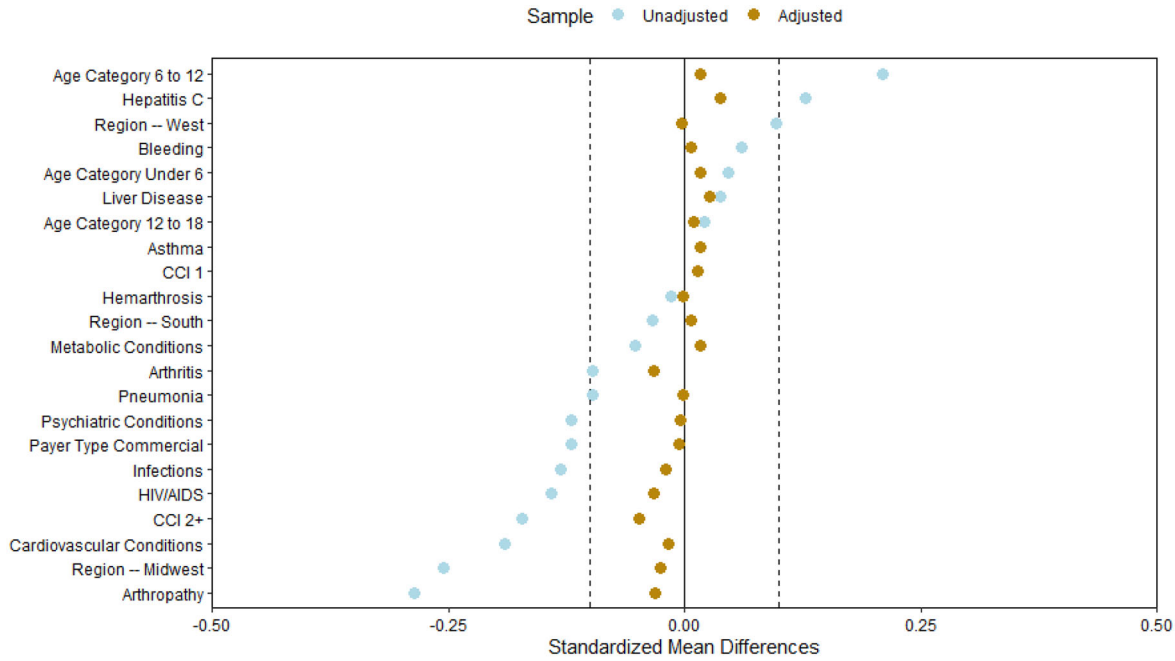

Figure 1. SMD of baseline clinical and demographic characteristics before and after propensity score weighting of patients in the octocog alfa and EHL groups. A positive SMD indicates a higher frequency in the octocog alfa group. An SMD with magnitude <0.1 (dashed line) indicates good group balance. Abbreviations. SMD, standardized mean difference.

survey (version 4.4-2) in R version 4.2.2. Outcome comparisons between weighted and unweighted concomitant octocog alfa and EHL FVIII cohorts were performed using independent *t*-test.

ABR_b and HCRU outcomes across cohorts were also compared using weighted multivariable Poisson regression, and weighted Gaussian regression was applied to cost and other continuous outcomes to adjust for confounders.

A sensitivity analysis limiting cohorts to patients with at least one year of follow-up was conducted to assess the impact of patients with short follow-up (<1 year). Outcome comparisons between these unweighted sub-cohorts were also undertaken using independent *t*-test.

Results

Cohort baseline characteristics

A total of 1,282 patients with HA without inhibitors on emicizumab prophylaxis were identified during the study period who had received emicizumab for a median [IQR] duration of 630 [309, 1220] days at index.

Among these patients, 501 were identified to have concomitant prescriptions for octocog alfa or EHL FVIII during the study period. Among the 501 patients, 274 (54.69%) were prescribed octocog alfa and 227 (45.31%) were prescribed an EHL, including 31 who were prescribed efanesoctocog alfa (Figure 2).

Baseline demographic and clinical characteristics of patients on emicizumab prophylaxis with concomitant octocog alfa and EHL FVIII prescription are described in Table 1. Patients who received concomitant octocog alfa were younger compared to those in the comparator EHL group. Both cohorts consisted of mostly males (>98%) and more than half (>60%) were covered by Medicaid. The distribution of index year, infections, metabolic conditions, bleeding history, psychiatric conditions, and comorbidity index was relatively similar across patient groups. However, a higher proportion of patients receiving EHL had cardiovascular conditions (10.1%) compared to those receiving octocog alfa (5.1%).

The efanesoctocog alfa sub-cohort consisted entirely of male patients with 54.8% being covered by commercial payer. About 19.4% of patients in this group had a bleeding history, which was higher than that of the octocog alfa (10.6%) and EHL (8.8%) cohorts.

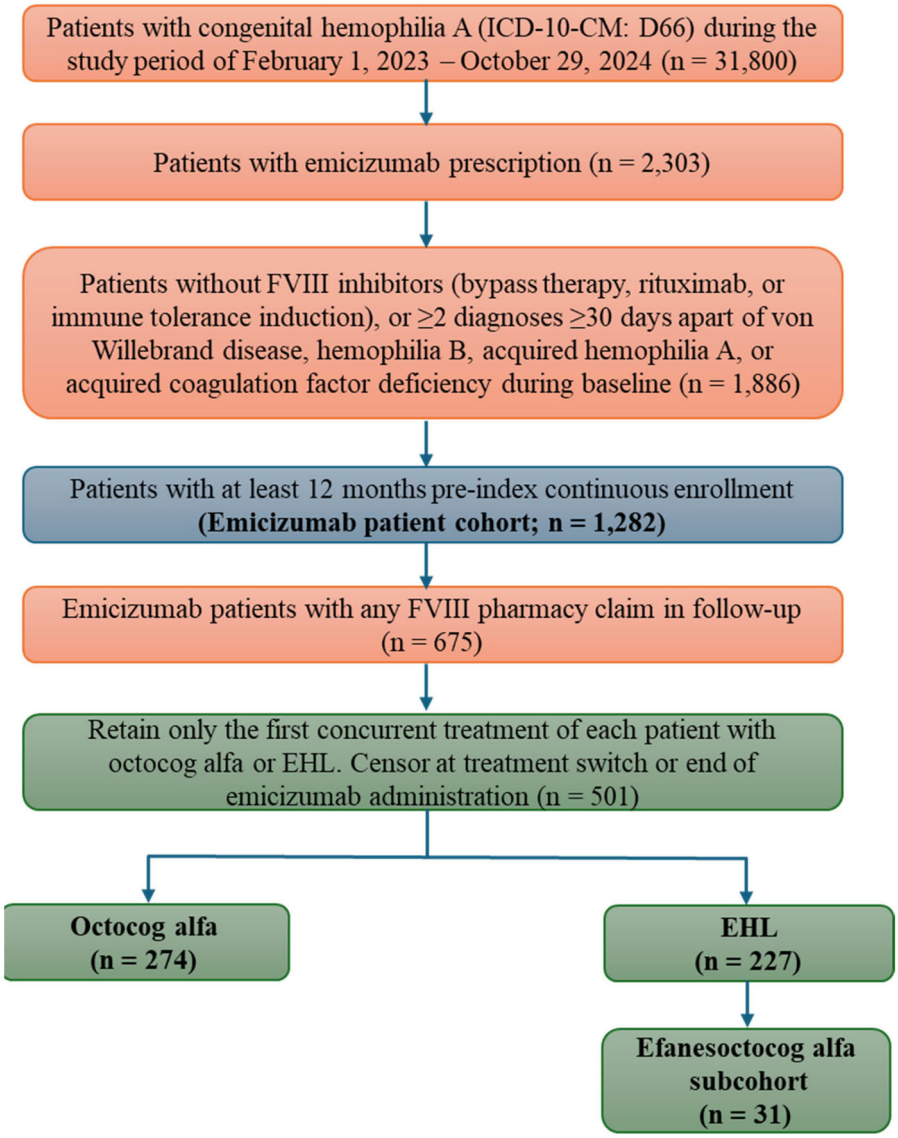

Figure 2. Patient selection flow diagram. The index claims of patients on emicizumab prophylaxis treated concurrently with EHL were most often efmoroctocog alfa ($n = 128$, 56%), followed by rurioctocog alfa pegol ($n = 53$, 23%), efanesoctocog alfa ($n = 31$, 14%), and damoctocog alfa pegol ($n = 15$, 7%). Note that although $n = 607$ patients did not have FVIII pharmacy claims during follow-up, they may still have received FVIII through medical-setting administration or from existing home supply. Patients with only medical FVIII claims were excluded from main analysis because HCPCS codes are not specific enough to reliably identify specific FVIII agent use. Abbreviations. ICD-10-CM, international classification of diseases, 10th revision, clinical modification; FVIII, factor VIII; EHL, extended half-life; HCPCS, healthcare common procedure coding system.

Study outcomes

The mean (SD) ABR_b among all patients receiving emicizumab prophylaxis was 0.28 (1.11). ABR_b was not different in a statistically significant manner across the concomitant octocog alfa and EHL cohorts with or without IPTW adjustment (Table 2 and Figure 3).

While FVIII utilization, all-cause ER, and outpatient visits were similar between the two cohorts, all-cause inpatient visits were significantly higher in the octocog alfa cohort than in the EHL cohort (p-value $= 0.03$) after IPTW adjustment. However, comparison of the bleed related inpatient visits did not show a significant difference across the two patient groups before and after IPTW adjustment. Bleed related outpatient visits also showcased a similar pattern (Table 2).

Table 1. Baseline characteristics of patients with HA without inhibitors receiving emicizumab prophylaxis and concomitant FVIII prescription.

Patients with emicizumab prophylaxis

Baseline characteristics	All patients (*N* = 1,282)	With concomitant FVIII prescription		
		Octocog alfa (*N* = 274)	EHL (*N* = 227)	Efanesoctocog alfa (EHL subgroup) (*N* = 31)
Demographic characteristics				
Age at index, median (IQR)	19.0 (18.0)	17.0 (14.0)	20.0 (15.5)	17.0 (23.0)
Age (years), *n* (%)				
<6	39 (3.0%)	8 (2.9%)	5 (2.2%)	1 (3.2%)
6–11	239 (18.6%)	62 (22.6%)	33 (14.5%)	4 (12.9%)
12–17	264 (20.6%)	70 (25.5%)	56 (24.7%)	11 (35.5%)
18+	740 (57.7%)	134 (48.9%)	133 (58.6%)	15 (48.4%)
Gender, *n* (%)				
Male	1268 (98.9%)	272 (99.3%)	224 (98.7%)	31 (100.0%)
Payer, *n* (%)				
Commercial	480 (37.4%)	80 (29.2%)	79 (34.8%)	17 (54.8%)
Medicaid	788 (61.5%)	191 (69.7%)	146 (64.3%)	14 (45.2%)
Medicare advantage	10 (0.8%)	3 (1.1%)	1 (0.4%)	0 (0.0%)
Unknown/missing	4 (0.3%)	0 (0.0%)	1 (0.4%)	0 (0.0%)
Region, *n* (%)				
Northeast	254 (19.8%)	39 (14.2%)	12 (5.3%)	2 (6.5%)
West	306 (23.9%)	78 (28.5%)	55 (24.2%)	7 (22.6%)
South	446 (34.8%)	91 (33.2%)	79 (34.8%)	8 (25.8%)
Midwest	276 (21.5%)	66 (24.1%)	81 (35.7%)	14 (45.2%)
Index year, *n* (%)				
2023	1212 (94.5%)	263 (96.0%)	222 (97.8%)	30 (96.8%)
2024	70 (5.5%)	11 (4.0%)	5 (2.2%)	1 (3.2%)
Clinical characteristics				
Infections total, *n* (%)	**153 (11.9%)**	**28 (10.2%)**	**33 (14.5%)**	**3 (9.7%)**
Hepatitis C	52 (4.1%)	14 (5.1%)	6 (2.6%)	0 (0.0%)
HIV/AIDS	48 (3.7%)	7 (2.6%)	12 (5.3%)	2 (6.5%)
Sepsis	3 (0.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Central line infection	4 (0.3%)	0 (0.0%)	1 (0.4%)	0 (0.0%)
Pneumonia	75 (5.9%)	14 (5.1%)	17 (7.5%)	1 (3.2%)
Metabolic conditions total, *n* (%)	**100 (7.8%)**	**18 (6.6%)**	**18 (7.9%)**	**3 (9.7%)**
Diabetes	38 (3.0%)	7 (2.6%)	5 (2.2%)	0 (0.0%)
Hyperlipidemia	71 (5.5%)	12 (4.4%)	14 (6.2%)	3 (9.7%)
Hypothyroidism	12 (0.9%)	3 (1.1%)	0 (0.0%)	0 (0.0%)
Cardiovascular conditions total, *n* (%)	**126 (9.8%)**	**14 (5.1%)**	**23 (10.1%)**	**3 (9.7%)**
Hypertension	110 (8.6%)	14 (5.1%)	18 (7.9%)	2 (6.5%)
Others[a]	38 (3.0%)	1 (0.4%)	7 (3.1%)	1 (3.2%)
Bleeding history total, *n* (%)	**132 (10.3%)**	**29 (10.6%)**	**20 (8.8%)**	**6 (19.4%)**
Major bleeding	15 (1.2%)	3 (1.1%)	4 (1.8%)	1 (3.2%)
Minor bleeding	119 (9.3%)	26 (9.5%)	17 (7.5%)	5 (16.1%)
Psychiatric conditions total, *n* (%)	**143 (11.2%)**	**29 (10.6%)**	**33 (14.5%)**	**6 (19.4%)**
Anxiety	134 (10.5%)	26 (9.5%)	29 (12.8%)	4 (12.9%)
Bipolar disorder	14 (1.1%)	4 (1.5%)	6 (2.6%)	2 (6.5%)
Depression	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Other total, *n* (%)	**442 (34.5%)**	**86 (31.4%)**	**78 (34.4%)**	**10 (32.3%)**
Liver disease	41 (3.2%)	9 (3.3%)	6 (2.6%)	0 (0.0%)
Arthritis	102 (8.0%)	12 (4.4%)	15 (6.6%)	2 (6.5%)
Arthropathy	172 (13.4%)	24 (8.8%)	42 (18.5%)	6 (19.4%)
Hematuria	38 (3.0%)	8 (2.9%)	2 (0.9%)	0 (0.0%)
Knee and hip replacement	29 (2.3%)	4 (1.5%)	5 (2.2%)	1 (3.2%)
Gastroesophageal reflux disease (GERD)	50 (3.9%)	11 (4.0%)	9 (4.0%)	2 (6.5%)
Hemarthrosis	108 (8.4%)	23 (8.4%)	20 (8.8%)	5 (16.1%)
Asthma	80 (6.2%)	18 (6.6%)	14 (6.2%)	1 (3.2%)
Respiratory insufficiency	32 (2.5%)	5 (1.8%)	4 (1.8%)	0 (0.0%)
Epilepsy	15 (1.2%)	3 (1.1%)	3 (1.3%)	0 (0.0%)
Osteoporosis/acute kidney failure/post hemorrhagic anemia/chronic obstructive pulmonary disease (COPD)	29 (2.3%)	0 (0.0%)	2 (0.9%)	1 (3.2%)
Charlson Comorbidity Index, mean (*SD*)	0.4 (1.1)	0.3 (0.9)	0.4 (1.0)	0.3 (0.6)
Charlson Comorbidity Index[b], *n* (%)				
CCI = 0	1056 (82.4%)	233 (85.0%)	183 (80.6%)	26 (83.9%)
CCI = 1	103 (8.0%)	24 (8.8%)	19 (8.4%)	2 (6.5%)
CCI ≥ 2	122 (9.5%)	17 (6.2%)	25 (11.0%)	3 (9.7%)

Abbreviations. FVIII, factor VIII; IQR, interquartile rate; EHL, extended half-life; HIV-AIDS, human immunodeficiency virus-acquired immunodeficiency syndrome; *SD*, standard deviation.

[a]Others include hemorrhagic stroke, peripheral vascular disease, deep vein thrombosis, heart failure, coronary artery disease (CAD), ischemic stroke, myocardial infarction.

[b]CCI = 0 indicates no significant comorbid conditions and lowest risk of mortality in a year; CCI = 1 indicates one mild comorbid condition with slightly increased risk of mortality; CCI ≥ 2 indicates multiple or more severe comorbidities with a higher risk of mortality.

Table 2. Comparison of outcomes between concomitant octocog alfa and EHL treatment groups among patients with HA without inhibitors receiving emicizumab prophylaxis.

Outcomes	Pre-IPTW			Post-IPTW		
	Octocog alfa ($n = 274$)	EHL ($n = 227$)	Unadj. *p*-value	Octocog alfa ($n = 274$)	EHL ($n = 227$)	Adj. *p*-value
Clinical, annualized mean (*SD*)						
ABR_b	0.35 (0.98)	0.28 (0.70)	0.36	0.35 (0.94)	0.28 (0.69)	0.34
HCRU all-cause, annualized mean (*SD*)						
ER visits	0.76 (1.90)	0.50 (1.06)	0.06	0.73 (1.76)	0.49 (1.04)	0.05
Inpatient visits	0.16 (0.82)	0.06 (0.28)	0.08	0.16 (0.8)	0.05 (0.26)	**0.03**
Outpatient visits	17.49 (37.46)	17.36 (37.59)	0.97	17.06 (36.48)	16.88 (37.02)	0.96
HCRU bleed-related, annualized mean (*SD*)						
ER visits bleed-related	0.09 (0.41)	0.05 (0.23)	0.32	0.09 (0.38)	0.06 (0.23)	0.17
Inpatient visits bleed-related	0.03 (0.17)	0.02 (0.14)	0.66	0.03 (0.17)	0.02 (0.13)	0.60
Outpatient visits bleed-related	0.61 (2.96)	0.49 (1.73)	0.58	0.59 (2.73)	0.45 (1.54)	0.46
FVIII utilization, annualized mean (*SD*)						
FVIII utilization (number of FVIII claims in follow-up)	2.48 (2.45)	2.81 (2.66)	0.16	2.51 (2.38)	2.69 (2.55)	0.42
Total cost of care (TCoC), annualized mean (*SD*)						
TCoC (Pharmacy + Medical) ($)	408,419.58 (289,148.18)	433,785.61 (246,080.67)	0.30	410,420.30 (278,998.40)	434,265.02 (256,153.65)	0.34
Pharmacy cost, annualized mean (*SD*)						
[a]Combined pharmacy cost (NDC + Jcode + Emicizumab) overall ($)	403,661.49 (288,712.44)	430,482.62 (245,649.76)	0.27	405,651.56 (278,800.94)	431,463.11 (255,826.40)	0.30
Emicizumab cost($)	368,461.32 (244,641.08)	364,969.24 (226,268.46)	0.87	370,270.82 (237,888.34)	369,723.85 (234,947.45)	0.98
[b]NDC/J code FVIII ($)	35,200.16 (71,651.84)	65,513.38 (90,067.06)	**<0.001**	35,380.74 (68,455.97)	61,739.27 (85,367.13)	**<0.001**
Medical cost, annualized mean (*SD*)						
[c]Medical cost (All-cause) ($)	1,253.51 (6,796.80)	897.78 (4,517.46)	0.50	1417.05 (7,226.30)	707.70 (3,850.98)	0.17

Abbreviations. IPTW, inverse probability of treatment weighting; EHL, extended half-life; *SD*, standard deviation; HCRU, healthcare resource utilization; FVIII, factor VIII; NDC, national drug code; ER, emergency room.
Bolded values represent statistical significance.
[a]Combined pharmacy cost defined as expenditures related to FVIII and emicizumab prescriptions, were identified using the relevant NDC and J codes.
[b]On demand concomitant FVIII identified *via* relevant NDC and J codes.
[c]Diagnostic costs were evaluated using medical claims data.

The total cost of care was similar between the patient groups receiving concomitant octocog alfa and EHL. However, prescription of concomitant octocog alfa showed a significantly ($p < 0.001$) lower annual mean (SD) pharmacy cost of $35,380.74 (68,455.97) compared with $61,739.27 (85,367.13) for concomitant EHL prescription, with or without IPTW adjustment (Figure 3). Similar trends were observed after adjustment for additional confounders (Table S2). Annual medical costs did not show a significant difference across the two cohorts pre and post IPTW adjustment.

The sensitivity analysis limiting cohorts to patients with at least one year of follow-up did not affect the significance of outcome comparisons between patients receiving concomitant octocog alfa and EHL, except that the previously observed higher all-cause inpatient visits in the concomitant octocog alfa cohort compared to the EHL group were no longer statistically significant (Table 3).

The annual mean (SD) pharmacy cost of concomitant efanesoctocog alfa prescription was $114,226.27 (107,099.38), which was significantly ($p < 0.01$) higher than the concomitant octocog alfa prescription cohort. A similar variation in annual pharmacy cost was observed between the concomitant octocog alfa and efanesoctocog alfa cohorts among patients $\geq$18 years of age ($49,355.12 *vs.* $ 179,155.97, $p < 0.01$).

Discussion

In this study conducted among patients with HA without inhibitors on emicizumab prophylaxis, octocog alfa was the most used concomitant FVIII agent (Table S4). At baseline, patients with a concomitant octocog alfa prescription were younger and less likely to have cardiovascular comorbidities compared with those receiving a concomitant EHL prescription.

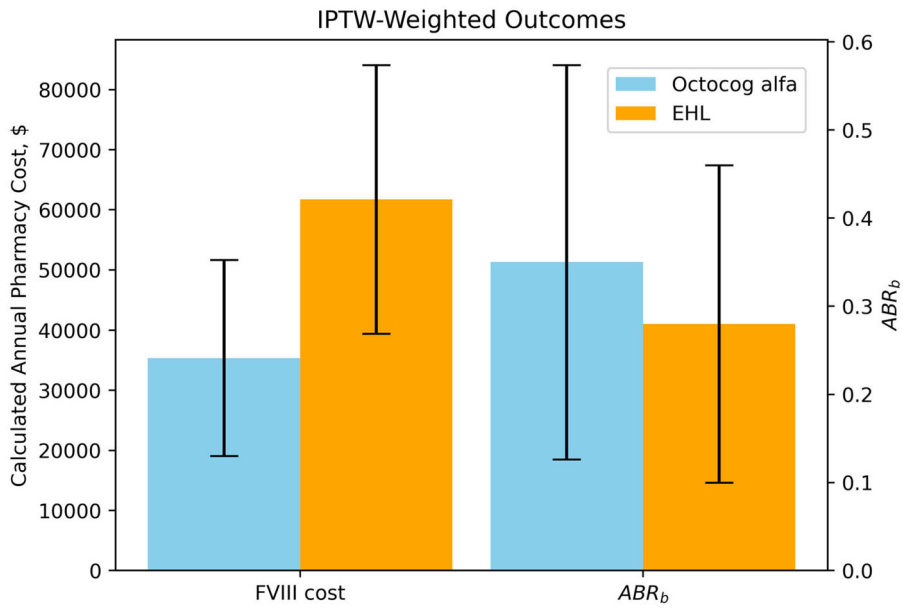

Figure 3. Calculated annual pharmacy costs of concomitant FVIII treatment and annualized bleeding rate (billable) among patients receiving emicizumab prophylaxis. Abbreviations. FVIII, factor VIII; EHL, extended half-life; IPTW, inverse probability of treatment weighting; ABRb, billable annualized bleeding rate.

Table 3. Comparison of outcomes between concomitant octocog alfa and EHL treatment groups among patients with HA without inhibitors with 1+ year of follow-up and receiving emicizumab prophylaxis.

	Full study cohort			1+ year follow-up		
	Octocog alfa ($n = 274$)	EHL ($n = 227$)	Unadj. *p*-value	Octocog alfa ($n = 171$)	EHL ($n = 140$)	Unadj. *p*-value
Clinical, mean (*SD*)						
ABR_b	0.35 (0.98)	0.28 (0.70)	0.36	0.32 (0.84)	0.30 (0.68)	0.79
HCRU all-cause, annualized mean (*SD*)						
ER visits	0.76 (1.90)	0.50 (1.06)	0.06	0.61 (1.06)	0.56 (1.02)	0.67
Inpatient visits	0.16 (0.82)	0.06 (0.28)	0.08	0.13 (0.68)	0.09 (0.33)	0.45
Outpatient visits	17.49 (37.46)	17.36 (37.59)	0.97	17.60 (38.12)	18.25 (39.73)	0.88
HCRU bleed-related, annualized mean (*SD*)						
ER visits, bleed-related	0.09 (0.41)	0.05 (0.23)	0.32	0.09 (0.28)	0.08 (0.26)	0.83
Inpatient visits, bleed-related	0.03 (0.17)	0.02 (0.14)	0.66	0.02 (0.13)	0.02 (0.11)	0.55
Outpatient visits, bleed-related	0.61 (2.96)	0.49 (1.73)	0.58	0.39 (1.03)	0.58 (1.98)	0.27
FVIII utilization, annualized mean (*SD*)						
FVIII utilization	2.48 (2.45)	2.81 (2.66)	0.16	1.87 (1.48)	2.22 (1.91)	0.07
Total cost of care (TCoC)**, annualized mean (*SD*)**						
TCoC (Pharmacy + Medical) ($)	408,419.58 (289,148.18)	433,785.61 (246,080.67)	0.30	375,851.35 (198,353.37)	398,014.32 (191,223.83)	0.32
Pharmacy cost, annualized mean (*SD*)						
Combined pharmacy cost (NDC + Jcode + Emicizumab) overall ($)	403,661.49 (288,712.44)	430,482.62 (245,649.76)	0.27	372,235.33 (198,983.75)	394,467.33 (190,297.55)	0.32
Emicizumab cost ($)	368,461.32 (244,641.08)	364,969.24 (226,268.46)	0.87	349,564.01 (190,724.31)	352,479.82 (181,875.37)	0.89
NDC/J code FVIII cost ($)	35,200.16 (71,651.84)	65,513.38 (90,067.06)	**<0.001**	22,671.32 (28,626.12)	41,987.51 (58,139.17)	**<0.001**
Medical cost, annualized mean (*SD*)						
Medical cost (All-cause) ($)	1,253.51 (6,796.80)	897.78 (4,517.46)	0.50	1,285.15 (6,415.51)	1,230.53 (5,392.34)	0.94

Abbreviations. IPTW, inverse probability of treatment weighting; EHL, extended half-life; *SD*, standard deviation; HCRU, healthcare resource utilization; FVIII, factor VIII; NDC, national drug code; ER, emergency room.
Bolded values represent statistical significance.

Our study reported that utilizing concomitant octocog alfa or EHL resulted in similar ABR_b, FVIII utilization, and HCRU (ER and outpatient visits) across both groups. Similar ABR_b among the two groups is expected as it is reflective of the real-world effectiveness of emicizumab prophylaxis rather than on-demand FVIII therapies. While all-cause inpatient visits were significantly higher in the concomitant

octocog alfa treatment cohort, the absence of a difference in bleed-related all-cause inpatient visits emphasized that the significant increase in all-cause inpatient visits were not bleed related. Furthermore, the lack of a difference in all-cause inpatient visits in the sensitivity analysis indicated that the significant difference observed in the full cohort was likely influenced by patients with shorter follow-up. This suggests that the choice of FVIII product, whether octocog alfa or an EHL agent, does not appear to substantially influence ABR_b and bleed related HCRU in terms of ER, inpatient, or outpatient visits when used alongside emicizumab.

The total cost of care and medical costs were similar between the groups, which is expected. Medical costs largely reflect disease control, which is primarily driven by emicizumab and is comparable across treatment groups according to the ABR_b endpoint. Pharmacy costs are also dominated by emicizumab, with annual costs 5–10 times greater than other FVIII therapies that may be used concomitantly. This suggests that medical costs and total pharmacy costs are not significantly influenced by the choice of FVIII product.

However, pharmacy costs associated with concomitant octocog alfa therapy were significantly lower than those associated with concomitant EHL prescriptions. The substantially lower pharmacy costs associated with octocog alfa highlight its potential for cost savings or minimization when used concomitantly with emicizumab, supporting its consideration as a preferred treatment option in appropriate patient populations. The findings of sensitivity analysis further validated the robustness of main analysis results. Our study further demonstrated that pharmacy costs associated with concomitant efanesoctocog alfa were significantly higher than those associated with prescription for concomitant octocog alfa.

A previous study reported a mean annualized ABR_b of 0.8 in patients with hemophilia A receiving emicizumab prophylaxis[15], which exceeds the value observed in the present study. Another study conducted among patients with HA without inhibitors receiving emicizumab prophylaxis reported an annualized mean cost of $50,491 for concomitant FVIII therapy during the first year of treatment. During the maintenance phase (>2 months follow-up), the study also reported an average per patient per month outpatient and ER visit rates of 1.55 and 0.03, respectively[17]. These estimates are consistent with the findings of our study, which reported mean annual costs of $35,380.74 for concomitant octocog alfa and $61,739.27 for EHL products. When translated to annualized rates, the healthcare resource utilization observed in the prior study further supports the credibility of our findings. The slight variation observed in cost and utilization patterns may be explained by our study's stratified analysis by concomitant FVIII product type, in contrast to the aggregated reporting of concomitant FVIII therapies in the comparator study. This stratified approach enhances the validity and granularity of our findings, providing more nuanced insights into the economic and clinical impact of emicizumab with concomitant FVIII therapy.

The use of octocog alfa in combination with emicizumab provides ABR_b comparable to those achieved with EHL FVIII agents, while offering a more cost-saving treatment option due to lower pharmacy costs. In the evolving HA landscape, considerations of safety, efficacy, treatment burden, and overall costs of care are important factors in deciding on individualized treatment. While the half-life extension is a notable difference between octocog alfa, EHLs, and efanesoctocog alfa, for patients already receiving prophylaxis with emicizumab, it is important to note that the majority of bleeds in clinical trials are resolved with 1–2 infusions of octocog alfa[30–33]. With FVIII peaks being the target in management of a bleed, the added half-life extension for a bleed that will resolve with one or two infusions of SHL FVIII comes at a significant cost to payers, over and above what they are already paying for emicizumab prophylaxis.

This study offers a valuable real-world perspective on treatment patterns and healthcare utilization among individuals with congenital HA without inhibitors. The use of a large, diverse claims database encompassing patients with Medicaid, Medicare, and commercial insurance enhances the generalizability of the findings to a broad population. Additionally, the application of standardized data cleaning and imputation algorithms, along with IPTW, strengthens the validity of comparative analyses by helping to reduce confounding bias.

Several limitations to this analysis should be acknowledged. As a retrospective claims-based study, the findings are subject to the typical constraints of administrative data, including potential coding

errors, omissions, and a lack of detailed clinical information. The study does not capture bleeds treated at home due to reliance on ICD-10 codes, leading to underestimation of ABR_b. Bleeds and the use of FVIII are not categorized according to cause, such as sports or post-trauma, due to lack of information. Patient weight is not captured, limiting the ability to assess its influence on dispensed quantity of FVIII, associated costs, or comorbidities. There is also potential for different components of a patient's therapy to be billed separately, with one product processed as a medical claim and another as a pharmacy claim, due to section 340B carve outs and related logistical challenges. As a result, treatments recorded only under the medical benefit may be underrepresented. Another limitation of the data was inability to isolate bleeding-related medical costs; however, this is partially mitigated by the separation of bleed-related HCRU from all-cause HCRU. Furthermore, the absence of data on disease severity, inhibitor status, and specific clinical outcomes limits the ability to assess the cost of illness or treatment effectiveness by clinical subgroups.

Assumptions made during the course of this analysis should also be acknowledged. Treatment exposure was determined using NDC codes from pharmacy claims data, which may not capture patients who received FVIII exclusively through medical settings. This could lead to an incomplete capture of patient with HA without inhibitors, however, most patients self-administer at home and are unlikely to receive treatment in only a medical setting. While J-codes were not used for patient identification of octocog alfa patients due to their non-specificity, they were utilized for calculation of FVIII utilization in the follow up. Conversely, pharmacy cost estimates may be subject to some inaccuracy, as the assumption is made that the FVIII that is being dispensed is being used, when in fact it may just be kept at the house and eventually expire; thus, there could be a potential overestimation of utilization and treatment-related costs. However, this limitation is expected to impact both patient groups similarly and is therefore unlikely to introduce substantial bias in comparative analyses.

Since efanesoctocog alfa was FDA approved in February 2023, and because of the nature of the closed claims database, this population was limited in the current study, and the smaller sample size allowed for descriptive analysis only. In addition, the study did not analyze SHL agents beyond octocog alfa. Switching off treatment was also assumed if the patient had at least one claim for an FVIII agent differing from their concomitant therapy at index. Lastly, despite the use of IPTW to address observable confounders, residual confounding from unmeasured variables may still be present.

Conclusion

In this study, octocog alfa emerged as the most commonly prescribed FVIII therapy for the management of breakthrough bleeds in patients with HA without inhibitors receiving emicizumab prophylaxis. When used as an adjunct to emicizumab in this patient population, octocog alfa provided comparable ABR_b to EHL FVIII, while being associated with significantly lower pharmacy costs. These findings indicate that octocog alfa may offer a cost-saving advantage over EHL FVIII products without compromising effectiveness. Further research is warranted, particularly to directly compare octocog alfa with efanesoctocog alfa.

Transparency

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Declaration of financial/other relationships

CM, JC, and MB are employees of Takeda Pharmaceuticals U.S.A., Inc., Lexington, MA, USA, and are Takeda shareholders. AC and AF are employees of Takeda Pharmaceuticals America, Inc., Cambridge, MA, USA, and are Takeda shareholders. EP is an employee of Axtria, Inc., Berkeley Heights, NJ, USA, a consulting company which was contracted by Takeda for statistical analysis of this study. The authors have no other relevant affiliations or financial

involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

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Author contributions

All authors contributed to the study conception, initial data analysis, and manuscript preparation. They were involved in drafting, reviewing, and revising the manuscript and approved the final version. All authors have given final approval of this version to be published and agree to be accountable for all aspects of this work.

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